

LESSON 12.5

VOLUME OF RIGHT CIRCULAR CYLINDERS – PART 2



LESSON INTRODUCTION

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

LEARNING TARGETS

- I can find the volume of a right circular cylinder.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

ESSENTIAL QUESTIONS

- How are radius and diameter related when finding the volume of a right circular cylinder?

LESSON NARRATIVE

The focus of this lesson is to find the volume of the right circular cylinder, given the measure of the radius or diameter and height of a cylinder. Students solve mathematical and real-world questions.

COMPONENT SUMMARY

12.5.1 WARM-UP (~5 MIN.)

Notice and wonder on farm silos

12.5.3 GUIDED PRACTICE (10 MIN.)

Practice calculating volume

12.5.5 YOUR TURN! (10 MIN.)

Independent practice of calculating volume

12.5.2 GUIDED INSTRUCTION (15 MIN.)

Review finding volume of cylinder

12.5.4 COLLABORATION (5 MIN.)

Error analysis calculating volume of cylinder

12.5.6 WRAP-UP (~5 MIN.)

Assessment of mastery of learning target

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Calculators
- (Optional) Chart Paper
- (Optional) Markers

ADDITIONAL PREPARATION

- 12.5.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle finding the area of the base of the cylinder when given the diameter.
- Students may struggle on questions in 12.5.2 Guided Instruction that provide circumference, but not the formula.

THINGS TO CONSIDER

- In questions that require several steps, students should not round until the final step as rounding mid-step could significantly change the answer. Consider using Desmos online scientific calculator.
- Consider having no-calculator questions where students turn their calculator over.
- If time is an issue, then consider assigning 12.5.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

12.5.2 Guided Instruction positions students to discuss the steps required to find the volume of a cylinder. Consider applying this routine for questions 2, 3, or 4. Students should answer the questions independently. Students can share their solutions in a low-stakes environment prior to sharing with the entire classroom.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Throughout the lesson, students will work on calculating volume of a cylinder using a few different formulas. To ensure mathematical fluency, students should pick accurate and efficient methods and formulas. Notice opportunities like question 3 in 12.5.2 Guided Instruction and question 2 in 12.5.4 Collaboration for students to highlight their process as much as or more than their product.

DIFFERENTIATE INSTRUCTION

In 12.5.3 Guided Practice provide an auditory entry-point by prompting students to discuss their strategies and answers with a partner. To provide additional scaffolding for struggling students, facilitate a teacher-led small group. Students can complete the statements individually then share, discuss, and critique within the small group.

12.5.1 WARM-UP: WHAT DO YOU WONDER?

QUESTIONING

Consider inviting an artist to join the class virtually or in person. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.



12.5.1 Warm-Up: What Do You Wonder?

1. A silo is a cylindrical container that is used on farms to hold large amounts of goods, such as grain. The artist Cam Scale painted a mural across 7 adjacent silos. The painting is of a young girl standing in an open meadow looking up into the open sky. The mural took about 4 weeks to paint.



- a. Describe at least two steps you think the artist needed to take before painting this mural.
Answers will vary. The artist had to find the surface area of the silos. The artist had to determine how much paint he would need.
- b. What is something you wonder about this artist's project?
Answers will vary. I wonder how the artist determined what scale to use for the girl. I wonder how the artist kept track of what to paint on each silo.



12.5.2 Guided Instruction: Finding the Volume of a Right Circular Cylinder

1. Complete the statements.

To find the volume of a cylinder,

☐ add
☒ multiply

the

☒ area
☐ circumference

of the base of the cylinder

☐ to
☒ by

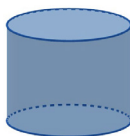
the height of the cylinder.

2. Landon says he can use the formula $V = \pi \left(\frac{d}{2}\right)^2 h$ to find the volume of a cylinder. Shayna disagrees and says the formula for finding the volume of a cylinder is $V = \pi r^2 h$. Discuss with your partner who you think is correct.

Answers will vary. Landon is correct because the diameter is half of the radius so by dividing by two, he is using the radius. Shayna is correct because the area of a circle is given by the formula πr^2 .

3. The cylinder shown has a diameter of 6 meters (m.) and a height of 4.8 m.

$d = 6 \text{ m.}$



$h = 4.8 \text{ m.}$

12.5.2 GUIDED INSTRUCTION: FINDING THE VOLUME OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

This lesson spirals content from Unit 4. Students may need to be directed back to this unit to remember some of the relationships they learned. Allow students to productively struggle in performing this research before giving them relationships from that unit that are used in this lesson. The struggle will benefit a student's learning and processing of the content.

After students share what they found, remember to facilitate a conversation about different formulas or methods for finding area of a circle: $A = \pi r^2$, $A = \pi \left(\frac{d}{2}\right)^2$, or $A = \frac{C^2}{4\pi}$. Since there are different ways to determine the area of a circle, then there will also be different ways to find the volume of a right circular cylinder.

TEACHER MOVES

Think-Pair-Share

Depending on your students' progress in Lesson 4, consider prompting students to engage in each question through the Think-Pair-Share routine for both individual and collective learning (MA.K12.MTR1.1).

- Students should work on each question individually and record answers.
- Students will pair with their partner or group members to discuss solutions and provide constructive critiques.
- The teacher will select a few groups to share their solutions and methods.

12.5.2 GUIDED INSTRUCTION: FINDING THE VOLUME OF A RIGHT CIRCULAR CYLINDER (CONTINUED)

DIFFERENTIATE INSTRUCTION

If your students learn best actively and collaboratively, consider structuring questions 3a-c as a Gallery Walk.

- Students can highlight and underline key terms.
- Students can annotate examples.
- Students can write key formulas on chart paper to be posted on classroom walls.

Prompt students to then circulate and visit the posters of their peers, using Mark-Up Text to annotate what they agree with, what they have questions about, or what they learned.

TEACHER MOVES

Ask students which formula they prefer and why. Prompt students to explain why the units are cubed.

QUESTIONING

Students who finish early may expand their thinking by calculating the volume of one or more of the cylinders. Answers can be left in terms of π or rounded to the nearest tenth.

DIFFERENTIATE INSTRUCTION

Students may struggle with how to use circumference without the formula. Facilitate a conversation about different formulas or methods for finding circumference of a circle: $C = 2\pi r$ OR $C = \pi d$.

- a. With your partner, decide who is partner A and who is partner B. Then calculate the volume of the cylinder using the formula indicated. Leave your answer in terms of π .

Partner A Use Landon's formula, $V = \pi \left(\frac{d}{2}\right)^2 h$	Partner B Use Shayna's formula, $V = \pi r^2 h$
$V = \pi \left(\frac{6}{2}\right)^2 (4.8) = 43.2\pi$	$V = \pi(3)^2(4.8) = 43.2\pi$

- b. Both formulas should result in the same answer. Check your answer with your partner. Then, discuss the similarities and differences in your calculations. Write a summary of your discussion.

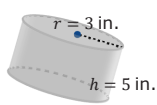
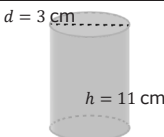
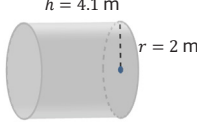
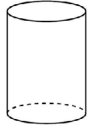
Answers will vary. One similarity is that both Landon and Shayna used π and the same height. One difference is that Landon used the diameter and Shayna used the radius in their formulas.

- c. Complete the statement.

The volume of the cylinder is 43.2π

- ☐ m.
☐ m^2 .
☒ m^3 .

4. Four cylinders are shown and specific measurements are given. The measurements include the radius, r , diameter, d , height, h , or circumference, C .
- a. For each cylinder, select which formula you would use to find the volume.

Cylinder	Formula
	☐ $V = \pi \left(\frac{d}{2}\right)^2 h$ ☒ $V = \pi r^2 h$
	☐ $V = \pi \left(\frac{d}{2}\right)^2 h$ ☒ $V = \pi r^2 h$
	☐ $V = \pi \left(\frac{d}{2}\right)^2 h$ ☒ $V = \pi r^2 h$
	☒ $V = \pi \left(\frac{d}{2}\right)^2 h$ ☐ $V = \pi r^2 h$

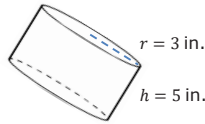
- b. Either formula can be used to calculate the volume of a cylinder. Compare your selections above with your partner. Discuss your reasoning for each selection.

Answers will vary. If you are given the diameter it is more convenient to use the formula $V = \pi \left(\frac{d}{2}\right)^2 h$. If you are given the radius it is more convenient to use the formula $V = \pi r^2 h$.

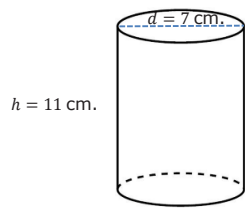


12.5.3 Guided Practice: Using Diameter or Radius to Find the Volume of a Right Circular Cylinder

1. Calculate the volume of each right circular cylinder. Leave your answers in terms of π .



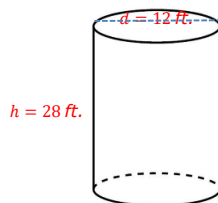
$$V = \pi(3^2)(5) = 45\pi \text{ in}^3$$



$$V = \pi \left(\frac{7}{2}\right)^2 (11) = \frac{539}{4} \pi \text{ cm}^3 = 134.75\pi \text{ cm}^3$$

2. On Mrs. Scarham's farm, there is a silo with a height of 28 feet (ft.) and diameter of 12 ft.

- a. Draw a sketch of the silo and label the dimensions.



- b. How many cubic feet of grain can this silo hold? Use 3.14 for π .

$$V = \pi \left(\frac{12}{2}\right)^2 (28) = 3165.12 \text{ ft}^3$$

12.5.3 GUIDED PRACTICE: USING DIAMETER OR RADIUS TO FIND THE VOLUME OF A RIGHT CIRCULAR CYLINDER

DIFFERENTIATE INSTRUCTION

Prompt students to make connections with previous lessons by creating partners or small groups.

- Consider grouping using 12.4.6 Wrap-Up data.
- Facilitate a teacher-led small group.
- Students can complete the statements individually, then share, discuss, and critique within the small group.

TEACHER MOVES

As you circulate, notice how students are determining the volume. For students who need further scaffolding, suggest creating a table with the row headers of radius, height, area of the base, and volume. By ordering their thinking, students can build up to finding the volume.

TEACHER MOVES

MA.K12.MTR.3.1: Mathematical Fluency

Facilitate conversations with students about mathematical fluency. Students should check their methods and calculations for reasonableness and accuracy. Students can discuss their solutions with partners to determine accuracy.

12.5.4 COLLABORATION: ERROR ANALYSIS

TEACHER MOVES

This error analysis shows one of the most common errors that students make. 12.5.6 Wrap-Up will show if students are carefully considering this error as they find the volume.

DIFFERENTIATE INSTRUCTION

For students who best demonstrate or express their understanding verbally, consider alternative ways to assess learning or understanding for this error analysis.

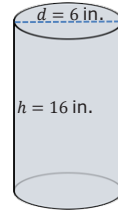
- Students could record themselves explaining the answer verbally.
- Students could use a chart to show the steps in a more visual manner.
- Students could tell a “story” or “narrative” of their interpretation of what happened. All forms can show an equal level of rigor and conceptual understanding.



12.5.4 Collaboration: Error Analysis

1. Simone calculated the volume of a cylinder with a diameter of 6 inches (in.) and height of 16 in. Her calculations are shown.

$$(\pi \cdot 6^2)(16) \\ 36\pi \cdot 16 \\ 576\pi \text{ in}^2$$



Simone made errors in her work. Work with your partner to determine what errors were made. Write a note to Simone identifying her errors and explaining how she can correct them.

Answers will vary. Simone made the mistake of using the diameter of the circle and with the radius of a circle's formula. She should have either used the formula using the diameter divided by two or cut the radius in half and then used that answer in the formula above. Also, her answer should be in cubic inches.

12.5.5 YOUR TURN!

TEACHER MOVES

Prompt students to find the volume of each cylinder prior to matching as a technique to answer question 1. Ask students to use formulas in previous activities to calculate the volume of each cylinder.

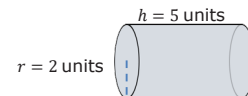
ENRICHMENT

- How do you know if the solution should be left in terms of π ?
- How do you ensure your solution is reasonable?



12.5.5 Your Turn!

1. The dimensions of four cylinders are given in the column on the left. Match each set of dimensions with the corresponding volume of the cylinder.
 1. **C.** $r = 2$ units; $h = 5$ units A. $36\pi \text{ units}^3$
 2. **D.** $r = 9$ units; $h = 3$ units B. $98\pi \text{ units}^3$
 3. **A.** $d = 6$ units; $h = 4$ units C. $20\pi \text{ units}^3$
 4. **B.** $d = 7$ units; $h = 8$ units D. Approximately 763 units^3
2. Find the volume of the cylinder shown. Use $\frac{22}{7}$ for π . Round your answer to the nearest thousandth.



$$V = \left(\frac{22}{7}\right)(2)^2(5) = 62.857 \text{ units}^3$$

3. The figure shows two grain storage silos. The diameter of each measures 24 feet (ft.) and the height of each cylinder measures 51 ft. Find the volume of the cylindrical portion of one of the silos. Use 3.14 for π .



$$V = (3.14)\left(\frac{24}{2}\right)^2(51) = 23060.16 \text{ ft}^3$$



12.5.6 Wrap-Up: Ticket Out the Door

- At a farm, animals are fed from buckets of grain. Each bucket of grain is a cylinder with a diameter of 12 inches. The height of the bucket is 14 inches. What is the volume of seed grain the bucket can hold? Use $\frac{22}{7}$ for π .



$$V = \left(\frac{22}{7}\right) \left(\frac{12}{2}\right)^2 (14) = 1,584 \text{ inches}^3$$

12.5.6 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

If many of your students make the error of using the diameter in place of the radius, then consider using student work and 12.5.4 Collaboration utilizing a My Favorite Mistake routine to draw students' attention to this very common error.

Another common error may be multiplying the fractions. Students could give the answer in any of the many equivalent forms. Consider further building student fluency by examining student work and leaving notes for students who are making computational errors with suggested improvements.

DIFFERENTIATE INSTRUCTION

For students who might struggle representing their understanding, consider alternative ways to assess or measure learning for this lesson.

- If technology is available, students could record themselves explaining the answer verbally.
- Students could provide detailed written methods of calculating volume of a cylinder and subsequently find the solution.
- Students could use a table or related aid to organize given details needed to find volume of the bucket.